

Fabrik Workflow Order & Inventory Management
System with Integrated SMS Notifications
for a
Custom T-Shirt Printing

A Capstone Project

Presented to the Faculty of the
College of Information Technology Education
PHINMA University of Iloilo

In Partial Fulfillment
of the Requirements for the degree
Bachelor of Science in Information Technology

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled "Fabrik Workflow Order&Inventory Management System with Integrated SMS Notification For a Custom T-shirt Printing" prepared and Submitted by John Rajeav Berame and Jasmar Quintar are hereby recommended for corresponding final evaluation.

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Abstract

The custom T-shirt printing industry is experiencing rapid growth driven by rising demand for personalized apparel. However, small businesses like FABRIK struggle with manual processes that cause order errors, inventory issues, pricing inconsistencies, and poor customer communication hindering efficiency and growth. This capstone project presents the FABRIK Workflow, an integrated Order and Inventory Management System (OIMS) with automated SMS notifications, tailored for small apparel businesses. The system centralizes operations, automates inventory updates, supports dynamic pricing and product management, and enhances customer engagement through real-time SMS updates. Key features include a robust order module, stock control panel, flexible pricing tools, transaction history and analytics. These tools aim to reduce manual errors, improve operational visibility, and support data-driven decisions. By addressing common inefficiencies through digital automation, the project contributes a scalable solution that improves workflow, boosts customer satisfaction, and supports sustainable growth. Limitations related to budget, training, and connectivity are considered, with a phased implementation strategy recommended.

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CHAPTER I

Introduction

The growing demand for personalized apparel has led to significant growth in the custom T-shirt printing market. This demand has led small businesses like FABRIK to play a key role in providing custom T-shirts for various occasions such as corporate events, personal celebrations, and retail. Despite the potential for growth, small businesses face several operational challenges that hinder their ability to scale efficiently. Issues such as manual order management, outdated inventory control systems, inconsistent pricing, and poor customer service often result in inefficiencies that affect profitability and competitiveness.

Fabrik has built a strong reputation for delivering personalized T-shirts tailored to customer specifications in terms of size, color, brand, print design, and delivery schedule. However, the company currently relies on manual, fragmented processes to manage its operations. Key tasks including order processing, inventory tracking, pricing, and production scheduling are performed without centralized systems, leading to a range of operational inefficiencies. These include order errors, inventory inconsistencies, pricing mistakes, and delayed deliveries, all of which negatively impact customer satisfaction and hinder the company's ability to scale effectively.

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The FABRIK Workflow aims to address these challenges, this study proposes the design and implementation of a centralized Order and Inventory Management System (OIMS) specifically tailored to Fabrik's business model. This system will unify and automate the company's core operational functions into a single digital platform. Key capabilities will include inventory tracking, order processing, dynamic and customizable pricing tools, and improved customer communication channels, including SMS notifications. The system is intended to reduce manual workloads, improve data accuracy, and support informed decision-making across departments. By enhancing internal efficiency and visibility, the OIMS will empower Fabrik to meet growing customer expectations and respond proactively to market fluctuations.

This transition to an automated and integrated system represents more than a technological upgrade, It is a strategic move to modernize Fabrik's operations, enhance service delivery, and position the company for sustainable growth. The implementation of the OIMS aligns with broader digital transformation trends and reflects Fabrik's commitment to innovation, customer satisfaction, and competitive agility in a fast-evolving marketplace.

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Background of the Study

Small businesses in the custom T-shirt printing industry, like FABRIK, often rely on manual processes for order management, inventory tracking, and pricing, which can lead to inefficiencies. These outdated methods result in problems such as inventory discrepancies, order fulfillment errors, and difficulty adapting to market changes. The FABRIK Production Tracker incorporates modern technologies to automate key business operations and improve overall efficiency.

The system uses the following key technologies:

- **Inventory Deduction,** As orders are processed, the system automatically deducts item quantities from the inventory, minimizing discrepancies and improving stock accuracy. This eliminates the need for manual stock updates and reduces the risk of overordering or stockouts.
- **Dynamic Product Attribute Management,** The system supports on-the-fly creation and modification of product attributes such as brand, size, type, and color without requiring system downtime or manual database adjustments.
- **Stock Control and Pricing Panel,** Administrators can easily manage stock levels and set flexible pricing based on variables such as order size, printing complexity, and material cost. This supports a more competitive and sustainable pricing model.

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- **Transaction Management and Order History,** The system captures detailed order information, including design previews, quantities, payment status, delivery timelines, and selected printing methods. This data ensures transparency and traceability for both customers and staff.
- **Automated SMS Notification System,** Customers receive timely updates through SMS, including order confirmation, production status, and delivery tracking. This enhances the customer experience by keeping them informed throughout the order process.
- **Receipt Generation,** Automatically generates digital or printable receipts after every confirmed transaction. This function ensures precise financial documentation, minimizes manual entry errors, and provides customers with official proof of purchase. Receipts include essential details such as order number, product summary, itemized pricing, applicable taxes, and payment confirmation—enhancing professionalism and reinforcing customer trust.
- **History and Reporting Module,** The system generates comprehensive historical data and reports on sales, inventory flow, and customer orders. These reports provide managers with valuable insights into business performance, helping them identify trends, forecast demand, and evaluate profitability. By having access to accurate historical records.

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Statement of the Problem

In the custom T-shirt printing industry, small businesses, such as FABRIK, encounter numerous operational challenges that impede their ability to meet customer demands, sustain profitability, and scale effectively. These challenges, primarily stemming from the reliance on manual processes and outdated systems, negatively impact critical business functions, including order management, inventory control, pricing strategies, and customer service. As a result, these inefficiencies hinder the growth potential of small enterprises in an increasingly competitive market.

1. **Manual Order Management,** The lack of an automated order management system leads to errors, delays, and costly rework, undermining customer satisfaction and retention.
2. **Inaccurate Inventory Tracking,** Without real-time tracking, FABRIK struggles with overstocking and stockouts, causing production delays, inefficient resource use, and cash flow disruptions.
3. **Static Pricing Mechanism,** Static pricing systems prevent FABRIK from adjusting prices in response to market fluctuations or customer demand, affecting competitiveness and profitability.
4. **Limited Transaction Visibility,** Without a centralized system, staff cannot easily access order histories, customer preferences, or material usage. This lack of visibility makes it difficult to track performance, manage

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workloads, and generate accurate reports—placing extra strain on employees, especially during peak periods.

5. **Poor Customer Communication,** Clients are not regularly updated on their order status due to the lack of automated notifications and tracking. Manual communication is slow and unreliable, leading to frustration and reduced trust in the service.
6. **Lack of Data-Driven Tools,** FABRIK lacks analytics tools to track sales trends, customer behavior, or inventory flow, making it hard to make informed decisions or adapt to changing demands.

This study proposes the development of an automated solution that integrates order management, inventory tracking, dynamic pricing, and enhanced customer service features. The goal is to enable small businesses like FABRIK to overcome these challenges and position themselves for long-term success.

This capstone project seeks to answer the following questions:

- **How can the system improve efficiency in order processing and tracking to minimize delays and reduce the risk of manual errors?**
- **How can inventory management be automated to ensure real-time monitoring of stock levels and prevent overstocking or shortages?**

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- How can the system standardize and automate pricing calculations based on product attributes such as size, color, quantity, and printing complexity?
- How can the system provide transparent transaction records and order histories for better data management and accountability?
- How can automated SMS notifications and receipt generation features improve communication, documentation, and customer trust?
- How can the system generate comprehensive reports to support business decision-making and performance evaluation?
- How can the system's user interface be designed to enhance usability, minimize training time, and promote efficient user interaction?
- How can the system be designed to handle scalability and adaptability, ensuring it can support business growth and future feature integration?

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Objectives of the Study

The primary objective of this study is to design and develop the FABRIK Workflow, a centralized Order and Inventory Management System (OIMS) with integrated SMS notifications a comprehensive solution that will:

- **Improve Order Processing Efficiency,** will develop an automated order management system that minimizes manual errors, accelerates processing time, and ensures accurate tracking of customer orders from placement to fulfillment.
- **Automate Inventory Management,** will implement real-time monitoring of stock levels to prevent overstocking or shortages, ensuring the accuracy of available materials and improving production planning.
- **Implement Dynamic Pricing Automation,** will design a pricing module that automatically adjusts costs based on product attributes such as size, color, quantity, and printing complexity to maintain competitiveness and profitability.
- **Ensure Transparent Transaction Recording,** will provide a centralized platform that stores and manages detailed order histories, transaction records, and payment information for easier access, traceability, and accountability.
- **Integrate Automated SMS Notifications and Receipt Generation,** will develop automated communication features that send customers real-time updates on their order status and generate official digital or printable receipts after each confirmed transaction to enhance professionalism and customer trust.

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- **Enhance System Usability and Accessibility,** will design a simple, intuitive, and user-friendly interface tailored specifically for FABRIK staff to efficiently manage daily operations such as order processing, inventory tracking, receipt generation, and reporting. The system will minimize training requirements, reduce human errors, and ensure that only authorized personnel can access and manage data securely.
- **Generate Comprehensive Business Reports,** will enable the system to produce detailed reports on sales, inventory movement, and customer activity to support data-driven decision-making, performance evaluation, and business forecasting.
- **Support Sustainable and Scalable Operations,** will develop a flexible and scalable system architecture that can adapt to future business growth or be customized for similar small enterprises within the apparel printing industry.

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Significance of the Study

The proposed FABRIK Workflow: Order and Inventory Management System (OIMS) with integrated SMS notifications is designed to address the operational inefficiencies commonly faced by small businesses in the custom T-shirt printing industry. As demand for personalized apparel continues to grow, small enterprises like FABRIK are under increasing pressure to streamline operations, improve accuracy, and deliver high-quality customer experiences. This study presents a comprehensive digital solution that enhances productivity, minimizes human error, and supports sustainable business growth.

At its core, the system automates the entire order management process—from order submission and design approval to production scheduling and fulfillment. This automation reduces manual workload, eliminates redundant data entry, and ensures timely and accurate delivery of customer orders. By improving process consistency and reliability, the system helps lower costs associated with reprints, delays, and operational inefficiencies, thereby improving both profitability and customer satisfaction. A vital component of the FABRIK OIMS is its real-time inventory tracking module, which automatically updates stock levels as transactions are processed. This feature eliminates the common issues of overstocking and shortages, improves inventory accuracy, and facilitates better forecasting and resource allocation. Managers can make data-driven purchasing decisions, reducing material waste and optimizing operational expenses.

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The dynamic pricing engine introduces flexibility and competitiveness by automatically adjusting prices based on various factors such as material cost, print complexity, order volume, and customer preferences. This ensures that pricing remains fair, transparent, and profitable, even in a fluctuating market environment. The integration of automated SMS notifications and digital receipt generation enhances the communication flow between FABRIK and its customers. Clients receive real-time updates regarding order confirmations, production progress, and delivery schedules, while receipts serve as official proof of purchase. These features strengthen customer trust, reduce the need for manual follow-ups, and promote a more professional image for the business.

The inclusion of data analytics and reporting tools enables comprehensive monitoring of business performance. Managers gain access to actionable insights on sales trends, order histories, inventory turnover, and customer behavior. These insights support strategic planning, operational optimization, and long-term decision-making. The FABRIK Workflow represents a strategic and future-ready solution for small businesses seeking to embrace digital transformation. By integrating automation, real-time monitoring, and customer-focused communication tools into one platform, the system enhances operational efficiency, reduces costs, and fosters business growth. Furthermore, this project serves as a replicable model for other small and medium-sized enterprises aiming to modernize their processes, improve productivity, and strengthen customer relationships through technology-driven innovation.

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Scope and Limitations of the Study

This study focuses on developing and implementing the FABRIK Workflow Order and Inventory Management System (OIMS)—a centralized platform designed to improve operational efficiency in the custom T-shirt printing industry. The system integrates automated order processing, real-time inventory tracking, dynamic pricing, reporting, and SMS notifications to minimize errors, streamline workflows, and enhance productivity and customer satisfaction.

The system is intended for staff and management use, enabling them to record customer orders, track inventory, generate receipts, and manage transactions efficiently. Customers only provide their order details, which are then encoded by the staff to ensure accuracy and organized documentation. The scope is limited to FABRIK's T-shirt printing operations and serves as a prototype for similar small businesses transitioning from manual to automated systems. The project prioritizes functionality, accuracy, and usability over advanced integrations or AI features.

However, the study recognizes the following limitations:

- **Budget Constraints** - As the target users are small businesses, financial resources may restrict the immediate adoption of advanced features such as full third-party integrations (e.g., e-commerce platforms or payment gateways) or AI-based analytics. To address this, FABRIK

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will implement the system in phases, focusing first on essential functions while leaving room for future upgrades as resources allow.

- **Industry-Specific Configuration** - While the system is optimized for the T-shirt printing industry, adapting it to other sectors (e.g., food production, general retail) would require structural and functional adjustments. Nevertheless, the system's modular design allows flexibility for customization to suit various business models.
- **User Adoption and Technical Training** - Since the system replaces manual processes, some staff may initially struggle with the digital workflow. To ensure smooth adoption, user training and documentation will be provided during the implementation phase.
- **Connectivity Dependence** - Since the system relies on cloud access and real-time updates, stable internet connectivity is essential for optimal performance. Areas with inconsistent or poor internet access may experience delays or functionality limitations.
- **SMS Notification Limitations** - SMS functionality is dependent on local network availability and may be subject to limitations based on the telecom provider's coverage or message queuing delays during peak hours. International SMS delivery may also be restricted or more costly.

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CHAPTER II

Literature Review

Overview of Electronic Marketing

Electronic marketing, also known as e-marketing or online marketing, involves using digital channels such as the internet, social media, email, and mobile apps to promote products and services. According to Efraim Turban and David King (2000) It has become a key strategy for businesses of all sizes, enabling them to reach a global audience while maintaining cost efficiency. Over time, the shift from traditional marketing methods to digital marketing has gained momentum, driven by the increasing importance of internet-based techniques. Today, businesses must adopt digital platforms to stay competitive and align their strategies with the evolving technological landscape, ensuring they can effectively engage with customers in an increasingly digital world.

Systematic Review

Systematic Review how electronic marketing impacts small businesses, Found evidence "the customers purchasing procedure and review the available aids that support the procedure", Consumer-oriented electronic marketing is also growing offline, mainly using smart cards, although it is still

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experimental. There are many common features between consumer-oriented and business-oriented marketing. Today the main concern of the management is shifting to how to utilize the opportunity of internet-based marketing to enhance competitiveness in harmony with existing marketing strategies. According to Efraim Turban and David King (2000), consumer purchasing activities can be broken down into seven key steps. These steps are critical for businesses to understand in order to meet customer needs and enhance their shopping experience.

1. Preliminary requirement determination
2. Search for the available items that can meet the requirement
3. Compare the candidate item with multiple perspective: specification, price, delivery date, and others terms or conditions
4. Place an order
5. Pay the bill
6. Receive the delivery items and inspect them
7. Contact the vendor to get after-service and support, or to return if disappointed

Direct vs Indirect Marketing

Direct Marketing relies on its own product according to "Efraim Turban, Jae Lee, David King, H. Michael Chung (2000)" Direct marketing works by selling products directly to customers through online stores or telemarketing, with no intermediaries involved. Manufacturers may seek their own products on the

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Internet if their electronic stores have high visibility, Company and brand is too low or managing an independent server is not economically justifiable which can be the case with small companies. The way direct marketing works by selling products directly to customers through online stores or telemarketing, with no intermediaries involved. On the other hand, indirect marketing involves third-party intermediaries like e-malls. The need for manufacturers to sell their products on well-known third-party platforms arises when their own electronic stores are less visible or economically justifiable.

Case studies

This case study exemplifies how a company can use both direct and indirect marketing strategies to enhance its online presence and compete in the e-commerce industry. The lessons learned from XYZ Electronics reflect the insights shared in *Electronic Commerce: A Managerial Perspective* about the importance of balancing internal visibility with external platform exposure, particularly when managing limited resources in an increasingly digital marketplace.

Efraim Turban, Jae Lee, David King, H. Michael Chung (2000). *Electronic Commerce A Managerial Perspective*: Prentice Hall. ISBN 981-4058-34-3

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CHAPTER III

Methodology

Research Design

This study employs a developmental research design, focusing on the creation, implementation, and evaluation of the FABRIK Workflow Order and Inventory Management System (OIMS). The developmental approach is appropriate because it allows for the systematic design and iterative refinement of the system based on real-world feedback from FABRIK's operations.

The research process begins with an in-depth analysis of FABRIK's existing workflows, followed by system planning, design, development, and testing. Iterative evaluation is conducted at each stage to ensure that the solution effectively addresses identified operational challenges, including manual order processing, inaccurate inventory tracking, static pricing, and inefficient customer communication.

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Model Used

The Agile Software Development Life Cycle (SDLC) model is adopted due to its flexibility, adaptability to changing requirements, and emphasis on stakeholder collaboration. Agile is particularly suitable for the custom T-shirt printing industry, where client demands and production workflows can change rapidly.



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The Agile development process for the FABRIK OIMS consists of the following phases:

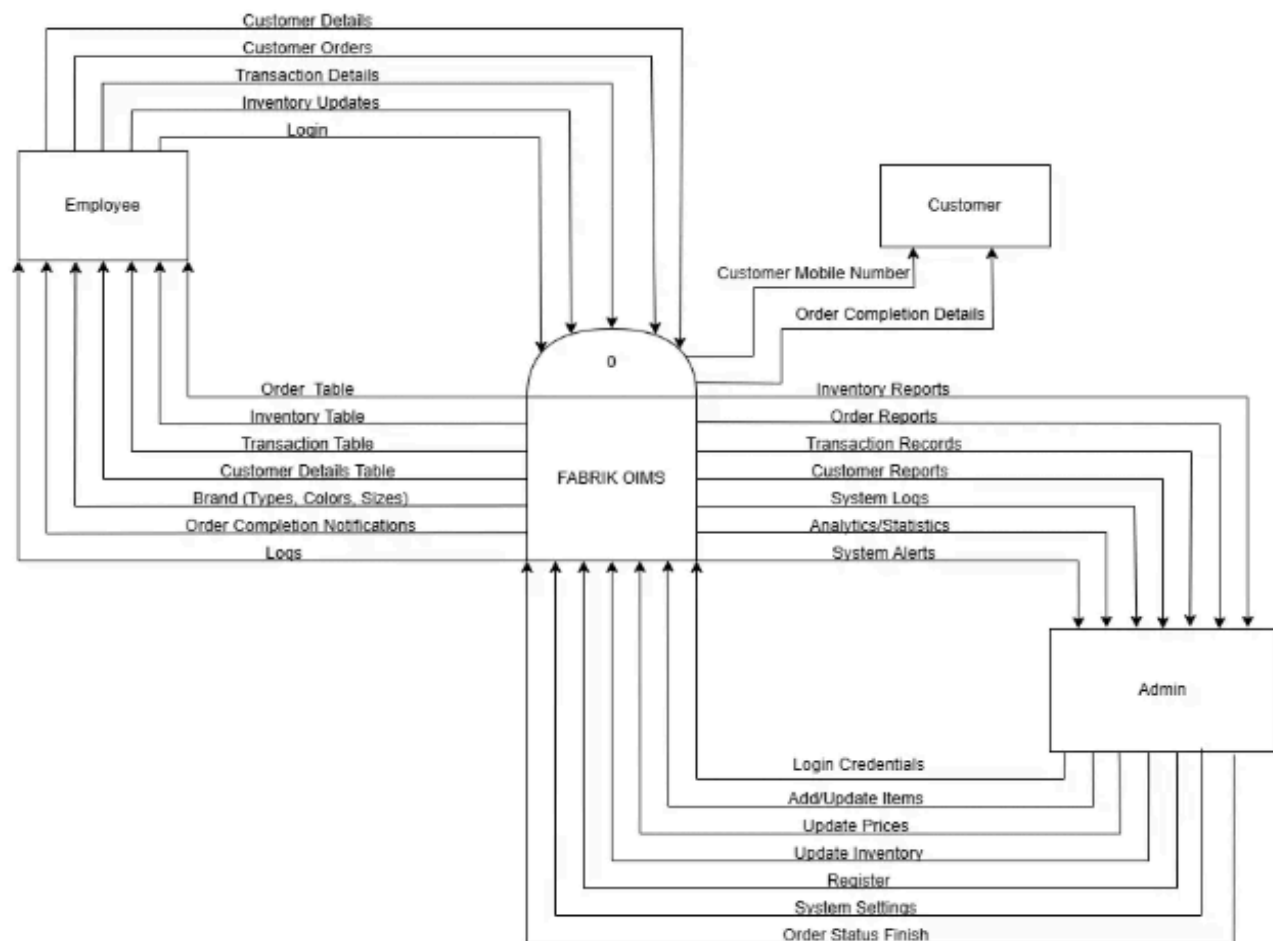
1. Planning and Requirement Analysis - Gathering system requirements through stakeholder interviews, workflow observation, and analysis of existing processes.
2. System Design - Creating user interface prototypes, database schemas, and scalable architecture to support real-time operations.
3. Development - Implementing system functionalities using appropriate technologies, including React.js for the front-end, PHP for the back-end, and MySQL for data management.
4. Testing - Conducting unit testing, integration testing, system testing, and user acceptance testing (UAT) to ensure accuracy, reliability, and usability.
5. Deployment - Launching the system in a live environment with user training sessions and technical documentation.
6. Maintenance and Support - Providing updates, bug fixes, and enhancements based on user feedback to ensure continued efficiency.

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Data Flow Diagram (DFD)

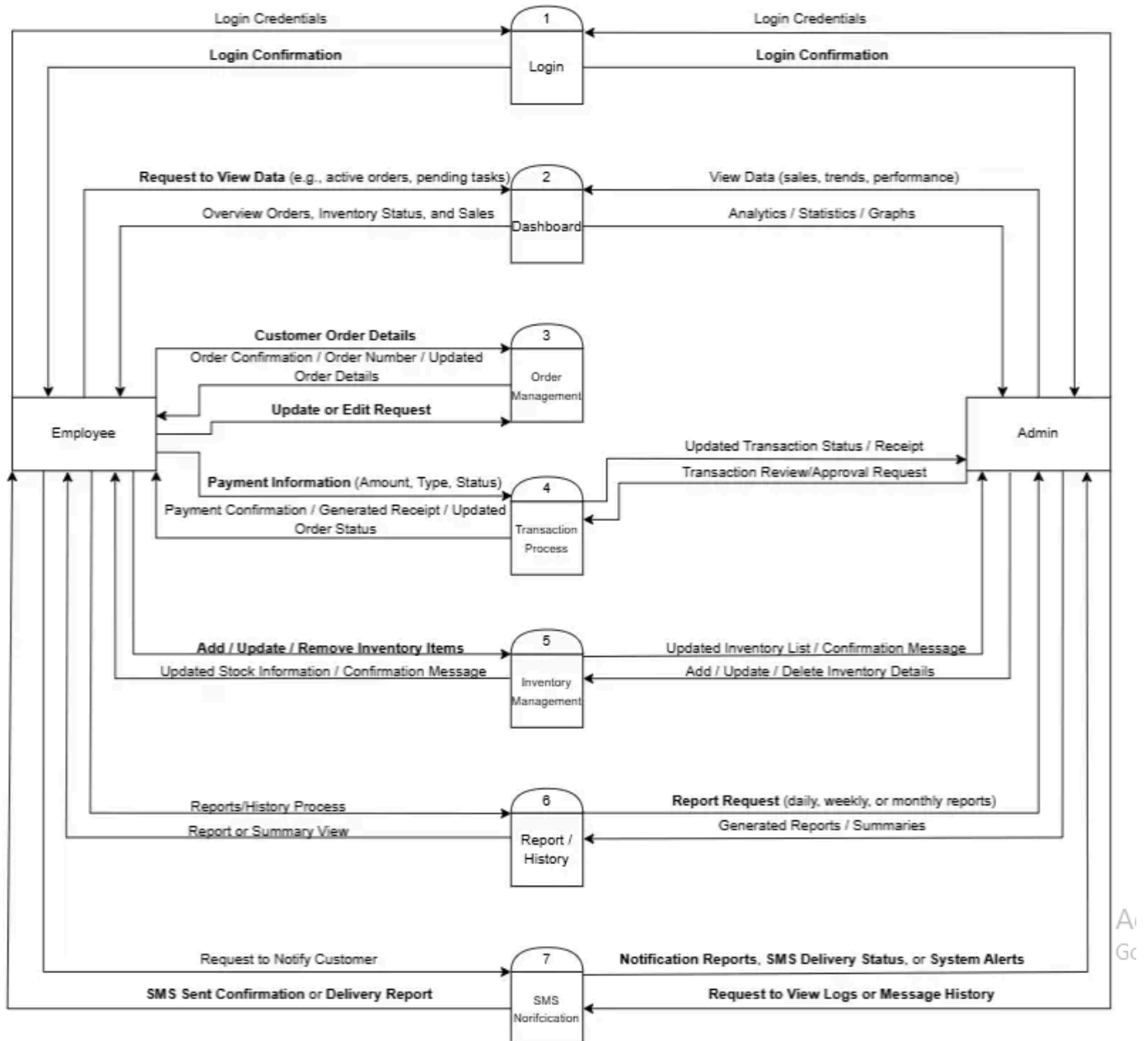
The Data Flow Diagram illustrates the logical flow of information within the FABRIK OIMS. It identifies the external entities, processes, data stores, and data flows that make up the system.

Level 0 DFD provides an overview of how data moves between customers, employees, admins, and the OIMS.



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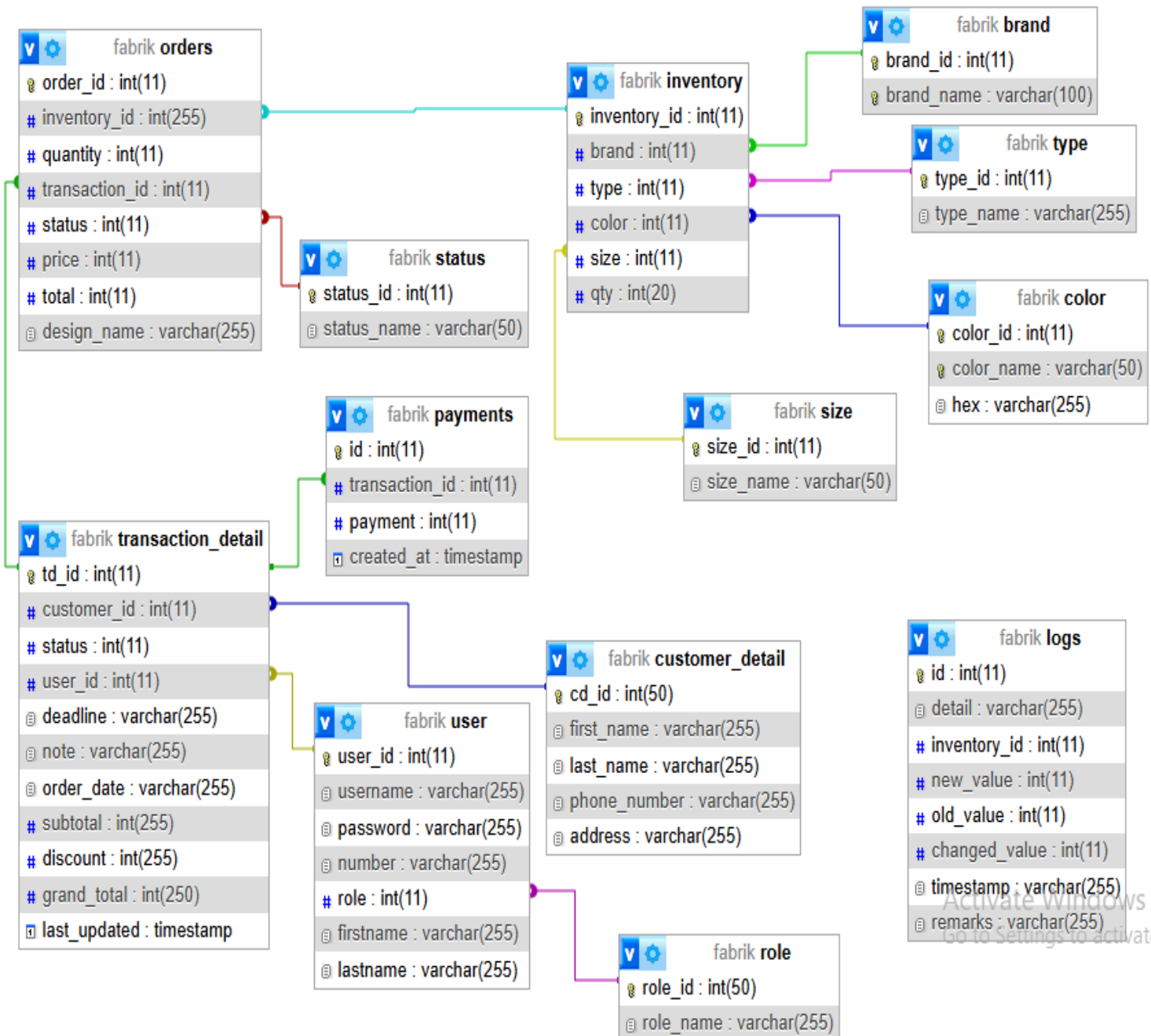
Level 1 DFD breaks down core processes such as order processing, inventory management, pricing adjustments, and SMS notification handling.



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Entity Relationship Diagram (ERD)

The Entity Relationship Diagram presents the logical structure of the database used in the FABRIK OIMS. It defines the relationships between entities such as:



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Data Gathering Procedures

Data collection is critical for understanding FABRIK's operational challenges and system requirements. The following methods are employed:

- **Interviews:** Conducting structured interviews with FABRIK's management, staff, and key stakeholders to gather insights into current workflows, pain points, and system expectations.
- **Observation:** Direct observation of existing processes in order management, inventory tracking, and pricing to identify inefficiencies and areas for improvement.

System Design and Development

The system design phase focuses on creating detailed technical blueprints for the FABRIK Production Tracker. This includes:

- **User Interface (UI) Design:** Developing intuitive, user-friendly interfaces for both staff and customers.
- **Database Design:** Structuring databases to support real-time inventory tracking, order management, and dynamic pricing.
- **System Architecture:** Designing a scalable, modular architecture that supports easy updates and integration with other business tools.

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Testing and Evaluation

Testing ensures the system operates as intended and meets all functional requirements.

- Unit Testing: Verifying individual components for correct functionality.
- Integration Testing: Ensuring that combined modules work together seamlessly.
- System Testing: Evaluating the complete system's performance under real-world conditions.

The FABRIK Production Tracker utilizes a range of tools and technologies to support development and deployment:

- Programming Languages: JavaScript, PHP and also HTML, CSS for web technologies in front-end and back-end development.
- Frameworks: React js for building dynamic user interfaces and Php for robust back-end development.
- Database: MySQL for secure, scalable data storage

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CHAPTER IV

RESULT AND DISCUSS

System's Features

The FABRIK Workflow Order and Inventory Management System (OIMS) was developed to automate and streamline the core operations of FABRIK's custom T-shirt printing business. The system integrates several key modules designed to improve efficiency, accuracy, and communication between staff and customers. Each feature addresses specific challenges identified in the manual process and contributes to a smoother, more organized workflow.

1. **Order Management** - This module allows staff to efficiently record and manage customer orders. Each order entry includes essential details such as customer information, T-shirt type, size, color, brand, quantity, and printing design.
2. **Inventory Tracking and Deduction** - The system automatically updates stock levels in real time, minimizing manual counting and preventing overstock or shortages. Staff can easily monitor quantities, add new stock, and detect low inventory to avoid production delays.
3. **Dynamic Pricing System** - The module automatically calculates total costs based on order quantity, material, and print complexity, ensuring consistent, transparent, and competitive pricing while protecting profit margins.

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4. **SMS Notification System** - The system automatically sends SMS updates for order confirmation, production status, and delivery, keeping customers informed, reducing inquiries, and building trust with clients.
5. **Receipt Generation** - The system automatically creates digital or printable receipts with order details, pricing, taxes, and payment confirmation, ensuring accurate records and providing customers with official proof of purchase.
6. **History and Reporting Module** - The system records all transactions, sales, and inventory activities, generating analytical reports that help track performance, identify trends, and forecast material needs for informed decision-making.

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Implementation Challenge

During the development and deployment of the FABRIK Workflow Order and Inventory Management System (OIMS), several implementation challenges were encountered that affected the project's progress and performance. These challenges emerged across the stages of system design, development, and user testing, and each required strategic solutions to ensure a successful outcome.

1. Technical Challenges, integrating multiple modules such as order processing, inventory tracking, and dynamic pricing into one centralized system required careful database synchronization and logic testing. During early stages, inconsistencies occurred between order data and inventory updates, leading to mismatched stock quantities. To resolve this, validation scripts and transaction logs were added to ensure real-time synchronization and data accuracy.
2. System Design and Interface Challenges, developing a user-friendly interface that minimized complexity while maintaining full functionality proved challenging. The system needed to cater primarily to staff with varying levels of technical knowledge. To address this, the design was simplified through intuitive navigation, organized dashboards, and labeled forms that reduced training requirements and input errors.
3. Resource and Time Constraints, limited resources both in terms of time and available development tools posed a challenge during implementation. Since FABRIK operates as a

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small enterprise, the project team prioritized core functionalities such as order tracking, receipt generation, and SMS notifications before expanding to optional features like advanced analytics or e-commerce integration.

4. **User Adaptation and Training**, transitioning from manual to automated operations initially created hesitation among staff accustomed to traditional processes. Some employees expressed uncertainty about using the system, particularly in handling digital records and tracking orders online. To overcome this, step-by-step guides and training sessions were conducted to build user confidence and promote system adoption.
5. **Connectivity and System Reliability**, as the system relies on internet-based functionalities, particularly for SMS notifications and real-time data updates, unstable connectivity occasionally interrupted transactions. These interruptions caused delays in order status updates and notifications. To mitigate this, the system was designed to queue SMS messages and automatically resend them once connectivity was restored.
6. **Data Validation and Security Measures**, ensuring data accuracy and security was another concern. Incorrect entries and accidental deletions could compromise reports and order records. To prevent this, user input validation and confirmation prompts were implemented, along with regular database backups to safeguard critical data.

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Summary of User Feedback on FABRIK OIMS

Aspect of Evaluated	Positive Feedback / Strength	Areas of improvement / Suggestion
Order Management	Automated processing reduced manual errors and improved accuracy.	Add a summary dashboard for faster overview of ongoing orders.
Inventory Tracking	Real-time updates helped monitor stock levels effectively.	Include visual alerts for low-stock or out-of-stock items.
Receipt Generation	Automatic receipts ensured accurate and professional documentation.	Option to customize receipt layout and details.
SMS Notification	Enhanced customer communication and reduced manual follow-ups.	Improve message timing and confirmation tracking.
System Usability	Easy-to-navigate interface; minimal training required.	Optimize page loading speed and visual design.

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Comparison to Other Systems

Before the implementation of the FABRIK Workflow Order and Inventory Management System (OIMS), FABRIK relied solely on manual processes for managing orders, inventory, pricing, and customer communication. These procedures involved handwritten records, basic spreadsheets, and verbal coordination among staff. While manageable at a small scale, this approach became inefficient and prone to human error as customer demand and production volume increased. Manual operations led to delays in order processing, inaccurate inventory records, and inconsistent pricing, which directly affected productivity, profitability, and customer satisfaction. The introduction of FABRIK OIMS transformed these manual workflows into a centralized, automated digital system that streamlined operations and enhanced data accuracy. Tasks that previously required hours of manual checking such as tracking available materials, updating stock levels, or preparing receipts are now completed automatically and in real time. The system's dynamic pricing, SMS notification, and receipt generation features not only improve efficiency but also strengthen communication and trust between the business and its customers.

Through automation, FABRIK OIMS eliminated redundant paperwork, reduced manual errors, and provided a transparent platform where staff can monitor order progress, stock availability, and transaction histories. This modernization has improved coordination between departments and ensured that decision-making is based on accurate, up-to-date information.

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Comparison between Manual Process and FABRIK OIMS

Features	Manual Process	FABRIK OIMS
Order Management	Orders recorded manually on paper or spreadsheets; prone to loss and duplication	Digital order recording with centralized database and real-time tracking
Inventory Control	Manual counting of stocks; frequent overstocking or shortages	Automated inventory deduction and real-time stock updates
Pricing Mechanism	Static and manually computed; error-prone	Dynamic pricing based on product attributes and order complexity
Customer Communication	Relies on manual texts or calls; often delayed	Automated SMS notifications and receipt generation ensure timely updates
Data Storage & Security	Paper-based or local files vulnerable to loss	Centralized and secure database with organized records
Reporting & Analytics	Time-consuming and inaccurate manual summaries	Instant system-generated reports and analytics for decision-making
Operational Efficiency	Slow, repetitive, and labor-intensive	Fast, automated, and streamlined processes with minimal supervision

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CHAPTER V

Conclusion

The development and implementation of the FABRIK Workflow: Order and Inventory Management System (OIMS) successfully addressed the key operational challenges faced by small businesses in the custom T-shirt printing industry. Prior to the system's introduction, FABRIK relied heavily on manual processes for order management, inventory tracking, and pricing, which often resulted in inefficiencies such as delays, data inaccuracies, and limited visibility of business operations. Through the integration of automated modules including order management, real-time inventory tracking, dynamic pricing, receipt generation, SMS notifications, and comprehensive reporting the system streamlined workflows, minimized manual errors, and enhanced data accuracy across all departments. As a result, FABRIK experienced improved operational efficiency, faster order fulfillment, and higher customer satisfaction.

The system provided management with reliable insights through its reporting and analytics features, allowing for more data-driven decision-making. This not only strengthened daily business operations but also positioned FABRIK for sustainable growth in a competitive market. Overall, the FABRIK OIMS has proven to be an effective digital solution for modernizing small-scale business operations, serving as a potential model for similar enterprises aiming to transition from manual to automated systems.

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Recommendation

To further enhance the effectiveness of the FABRIK Workflow Order and Inventory Management System (OIMS), it is recommended that continuous monitoring and evaluation be conducted to maintain system performance and reliability. Regular maintenance should be performed to ensure data accuracy, minimize technical issues, and sustain system efficiency over time. Staff should receive periodic training to reinforce familiarity with the system's features, especially for order processing, inventory updates, and receipt generation. Gathering user feedback from employees can help identify minor issues and guide future system enhancements.

The current version of the system already addresses major operational challenges, future updates may focus on improving the reporting and analytics modules, enhancing user interface design, and optimizing SMS notification delivery. Since FABRIK handles payments directly through in-person transactions, the inclusion of online payment features is unnecessary and outside the project's intended scope. Future developers are encouraged to explore system scalability and integration with other internal business tools—such as accounting or inventory forecasting systems—to further strengthen operational efficiency and business insights.

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Final Thoughts

The development and implementation of the FABRIK Workflow Order and Inventory Management System (OIMS) represent a major advancement in addressing the operational challenges commonly faced by small businesses in the custom T-shirt printing industry. By transitioning from manual to automated processes, the system showcases how technology can significantly improve business efficiency, accuracy, and productivity without requiring large-scale investments. Throughout the design and development process, the team focused on creating a practical solution that could be easily understood and operated by small business owners and staff. The system's automated order processing, real-time inventory tracking, and integrated receipt generation features not only reduce human error but also promote transparency and consistency in day-to-day operations. This advancement contributes to a smoother workflow, faster transactions, and a more organized business environment that ultimately enhances customer satisfaction.

Beyond the technical perspective, the project demonstrates the importance of innovation and adaptability in small business management. In an era where digital transformation defines competitiveness, the FABRIK OIMS serves as a concrete example of how automation can empower local enterprises to remain relevant and sustainable. It encourages small businesses to embrace modern tools that improve decision-making, enhance record accuracy, and strengthen customer trust through clear communication and documentation.

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The project highlights the value of teamwork, problem-solving, and continuous improvement in software development. Each challenge encountered during implementation provided valuable learning experiences that shaped the system into a reliable and user-centered platform. The success of this project reflects not only technical accomplishment but also the dedication and collaboration of the development team. The FABRIK OIMS stands as more than just a system it is a step toward innovation, efficiency, and growth. It demonstrates that even small businesses, with the right tools and mindset, can achieve digital transformation and long-term success in a competitive industry. This project serves as both a foundation and an inspiration for future developers to continue building systems that empower businesses through technology.

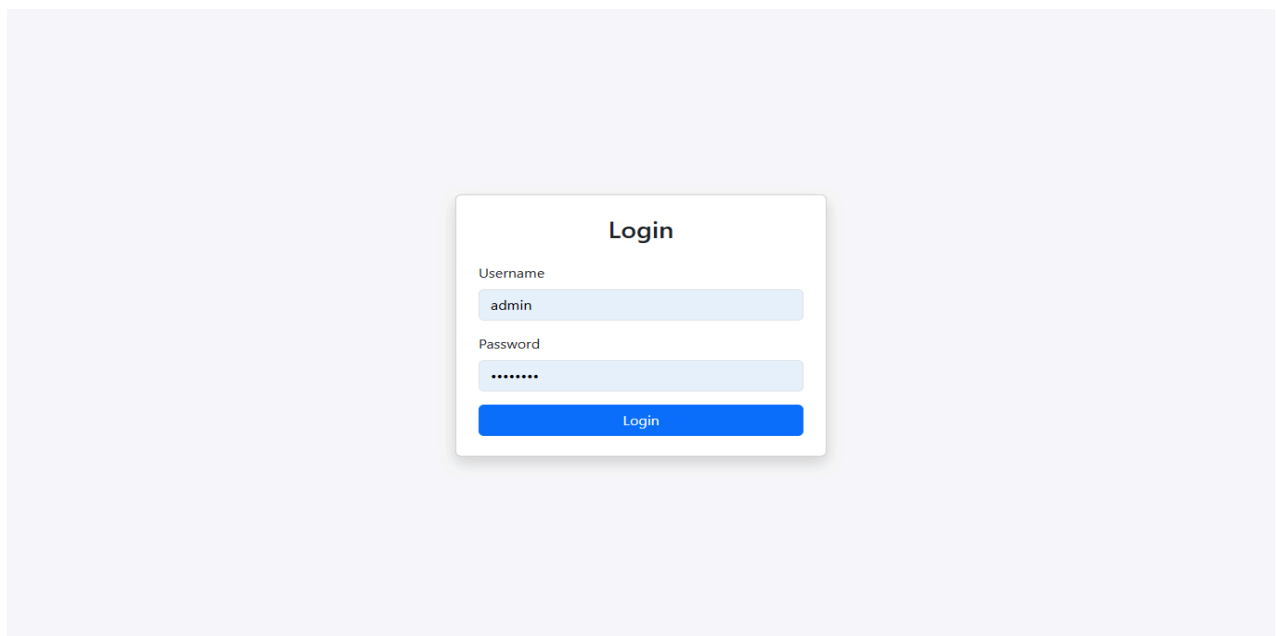
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Appendices

Screenshots

Login Page

Figure 1



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Dashboard Interface

Figure 2

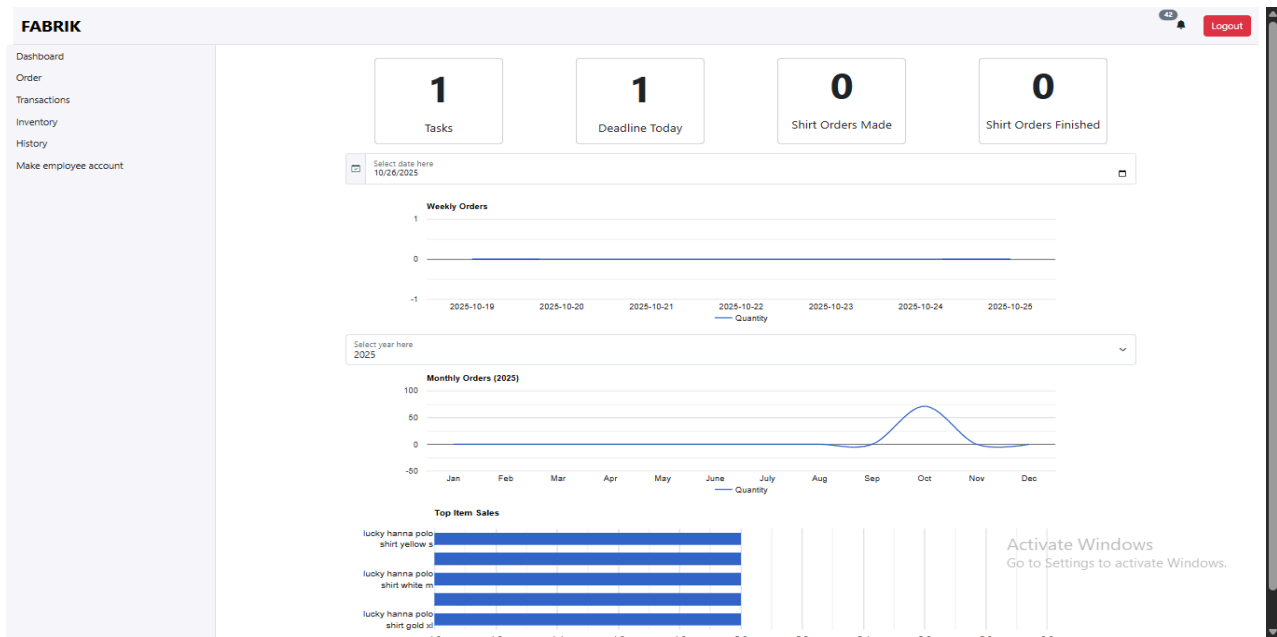
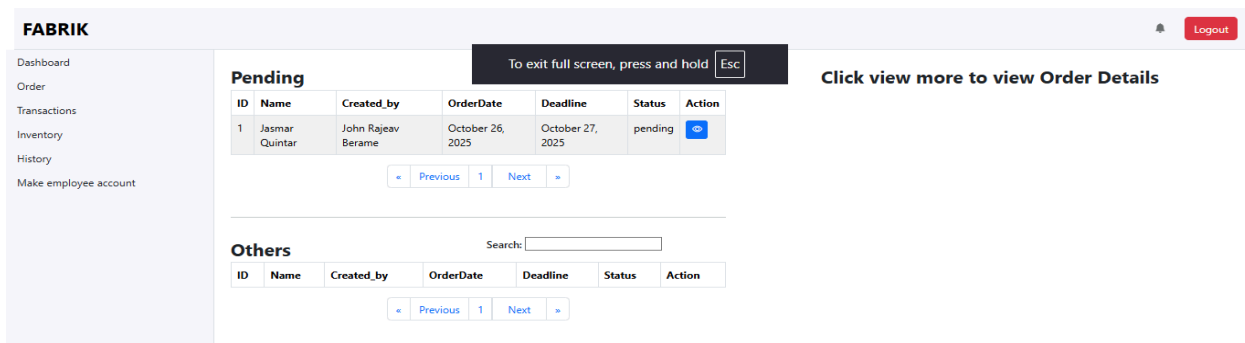


Figure 2.1



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Figure 2.2

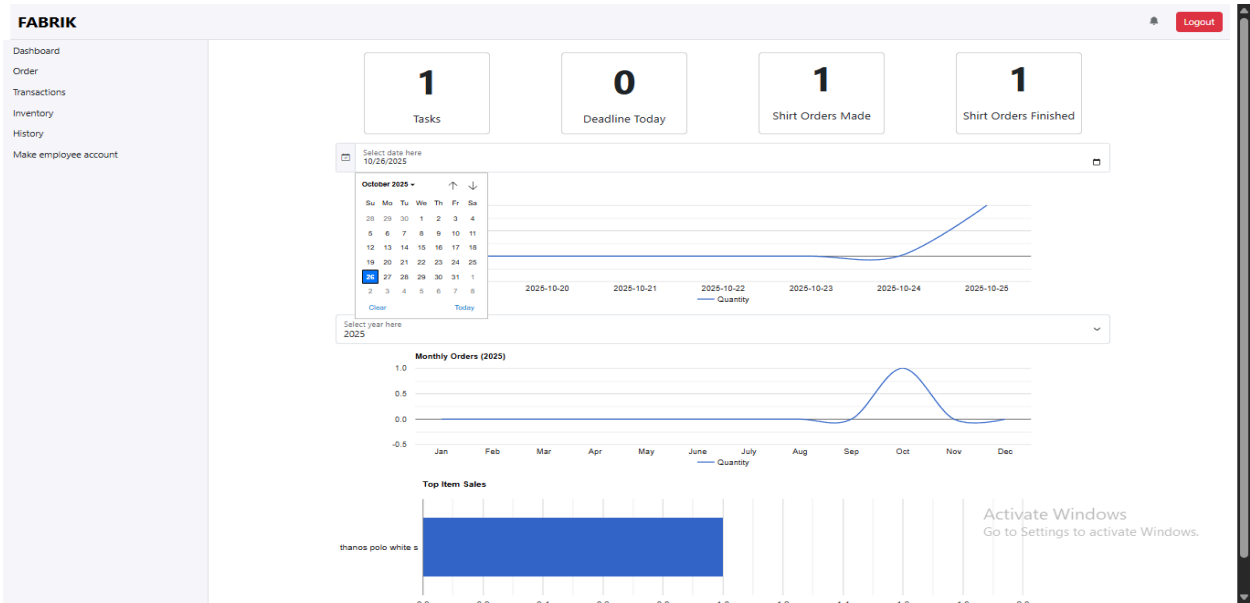
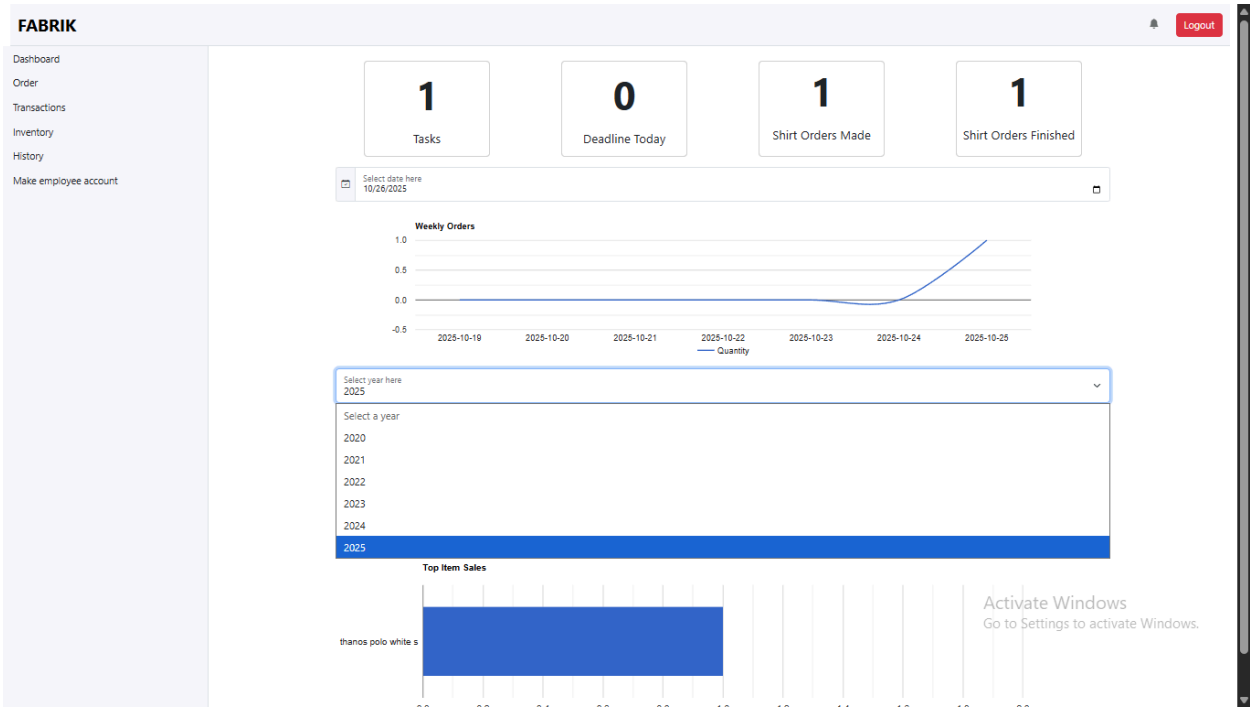


Figure 2.3



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Order Management Interface

Figure 3

FABRIK

Dashboard
Order
Transactions
Inventory
History
Make employee account

Brands
[Thanos](#)

Please select a brand

☐ Click here for quick order

Select Customer

Customer details

Firstname (required)

Lastname

Phonenumber (required)

Address

Transaction details

Deadline (required) mm/dd/yyyy

Design name

Note

Order	Quantity	Total	Action
Discount			
Grand total : 0			

Activate Windows
Go to Settings to activate Windows.

Figure 3.1

FABRIK

Dashboard
Order
Transactions
Inventory
History
Make employee account

Brands
[Thanos](#)

Types
Polo

Colors
White

Sizes
white

Size	Quantity	Action
s	2	Add Edit

☐ Click here for quick order

Select Customer

Customer details

Firstname (required) Jasmir

Lastname Quintar

Phonenumber (required) 09515573050

Address Near at Asian College Of Aeronautics, Ungka

Transaction details

Deadline (required) 10/28/2025

Design name

Note

Order	Quantity	Total	Action
Discount			
Grand total : 0			

Activate Windows
Go to Settings to activate Windows.

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Figure 3.2

FABRIK

Dashboard
Order
Transactions
Inventory
History
Make employee account

Brands
Thanos

Types
Polo

Colors
White

Sizes
S

white

Quantity: 2
Price: 1000

Close Save changes

Click here for quick order

Select Customer

Customer details

Firstname (required)
Jasmar

Lastname
Quintar

Phonenumber (required)
09515573050

Address
Near at Asian College Of Aeronautics, Ungka

Transaction details

Deadline (required)
10/28/2025

Design name

Note

Order	Quantity	Total	Action
Thanos Polo White S	2	2000	

Grand total : 0

Activate Windows
Go to Settings to activate Windows.

Figure 3.3

FABRIK

Dashboard
Order
Transactions
Inventory
History
Make employee account

Brands
Thanos

Types
Polo

Colors
White

Sizes
S

white

Quantity: 2
Price: 1000

Close Save changes

Click here for quick order

Select Customer

Customer details

Firstname (required)
Jasmar

Lastname
Quintar

Phonenumber (required)
09515573050

Address
Near at Asian College Of Aeronautics, Ungka

Transaction details

Deadline (required)
10/30/2025

Design name

Note

Order	Quantity	Total	Action
Thanos Polo White S	2	2000	

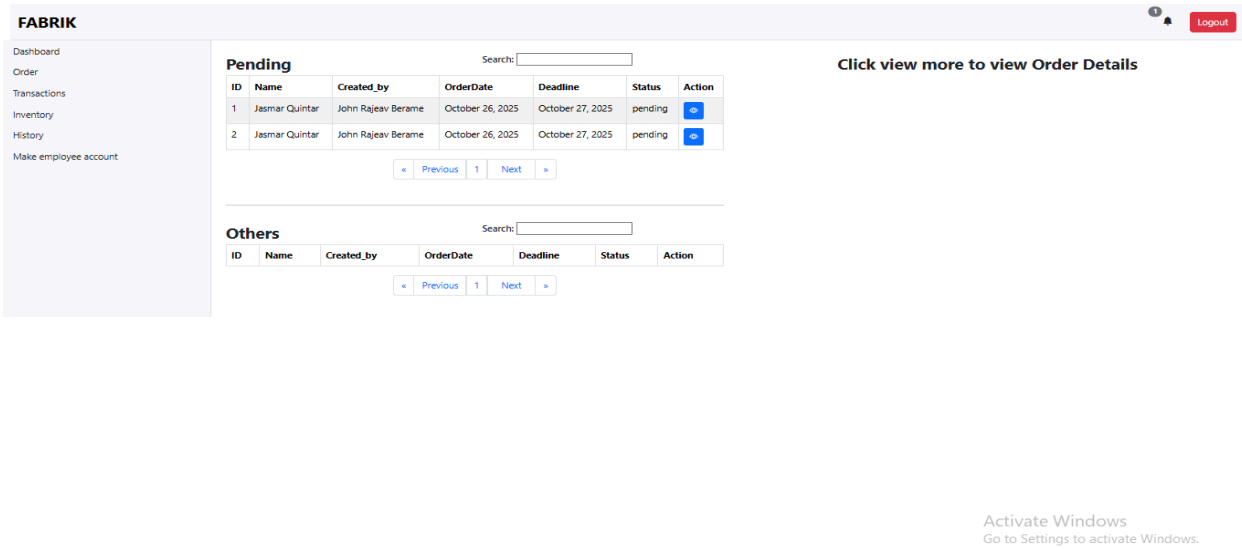
Grand total: 2000

Submit Order Cancel Order

Activate Windows
Go to Settings to activate Windows.

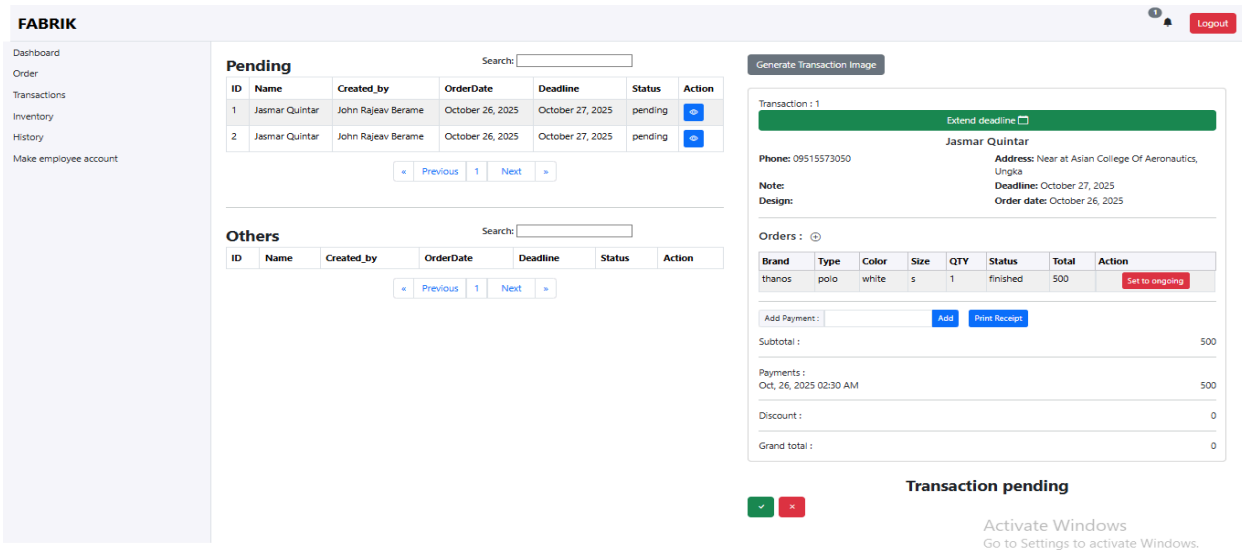
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Figure 3.4



Transactions Management Page

Figure 4



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Figure 4.1

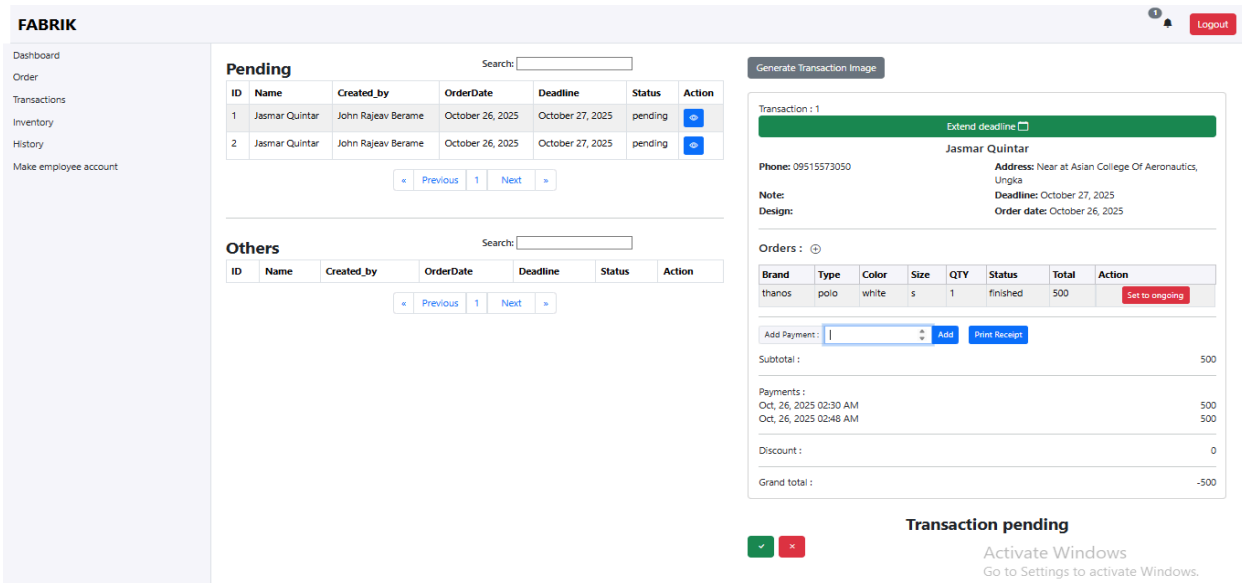
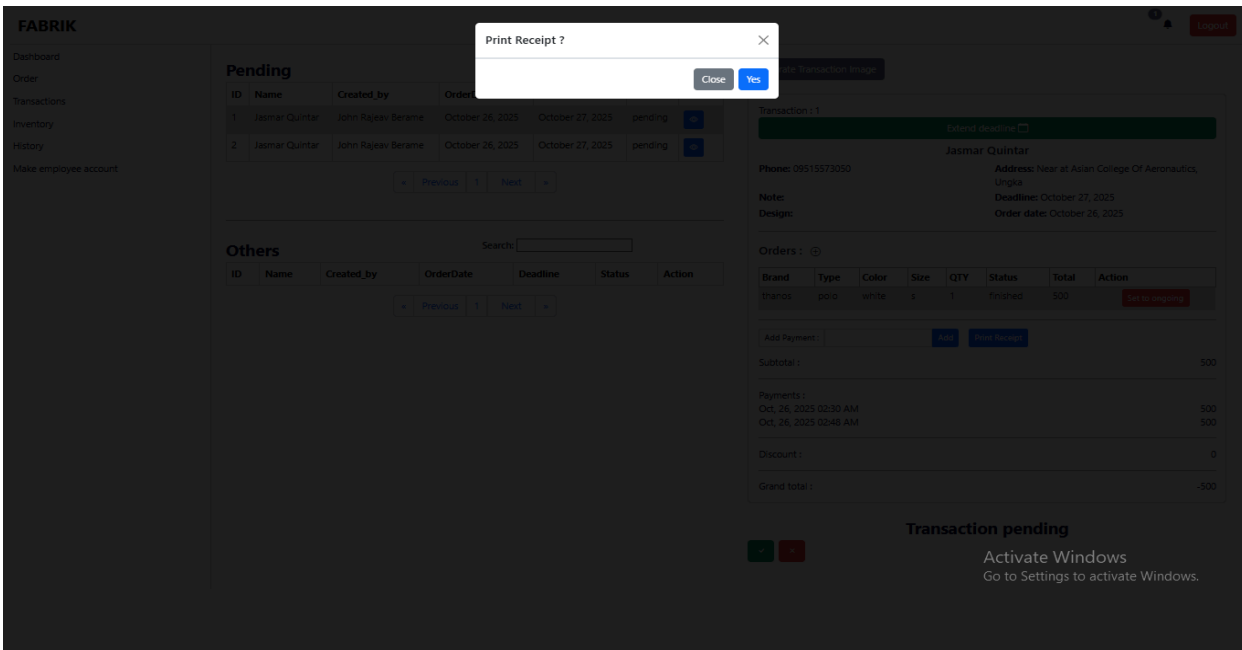


Figure 4.2



PHINMA UNIVERSITY OF ILOILO
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Figure 4.3

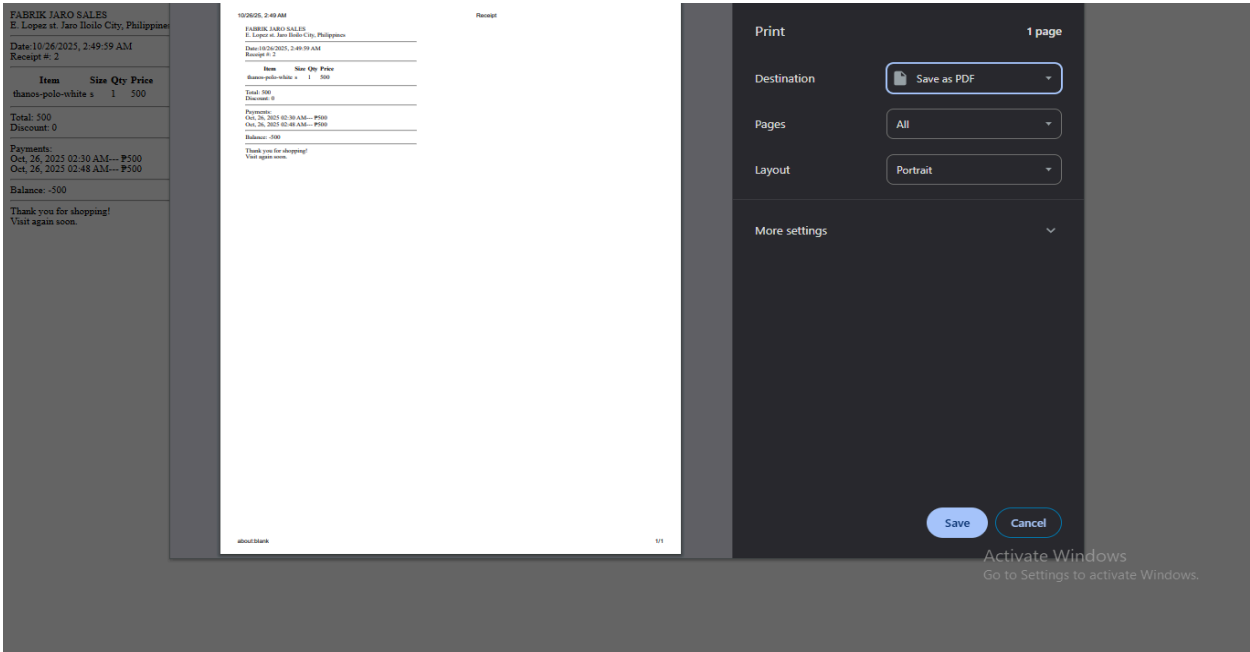
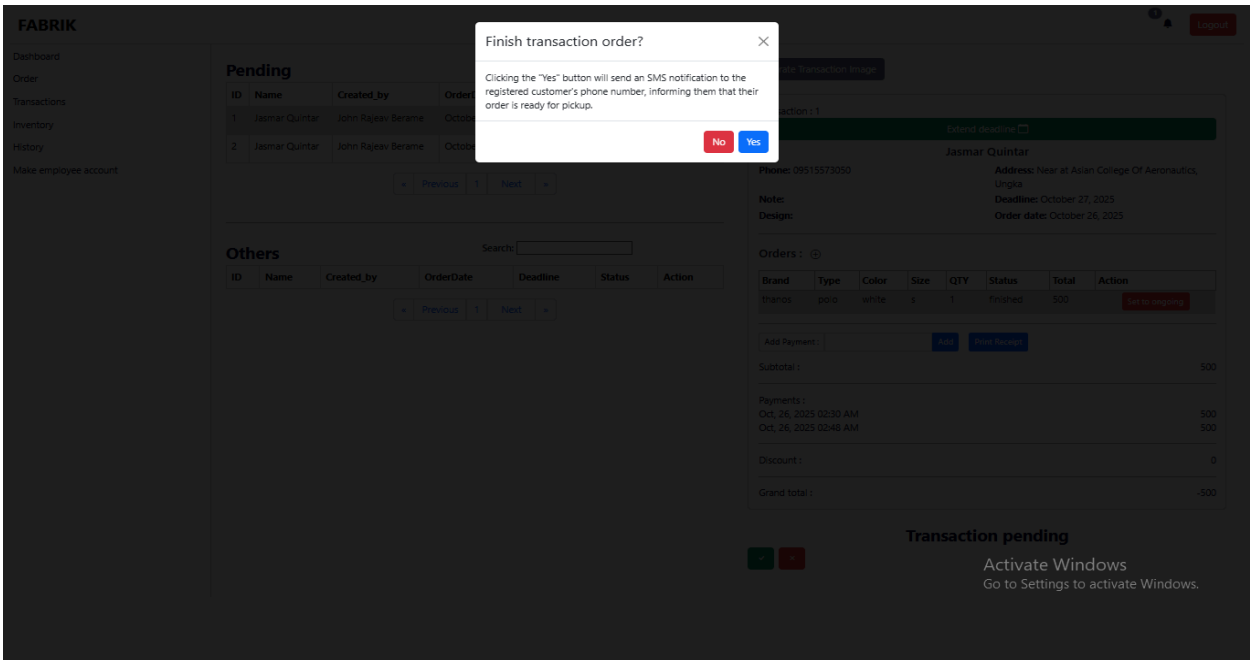


Figure 4.4



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Figure 4.5

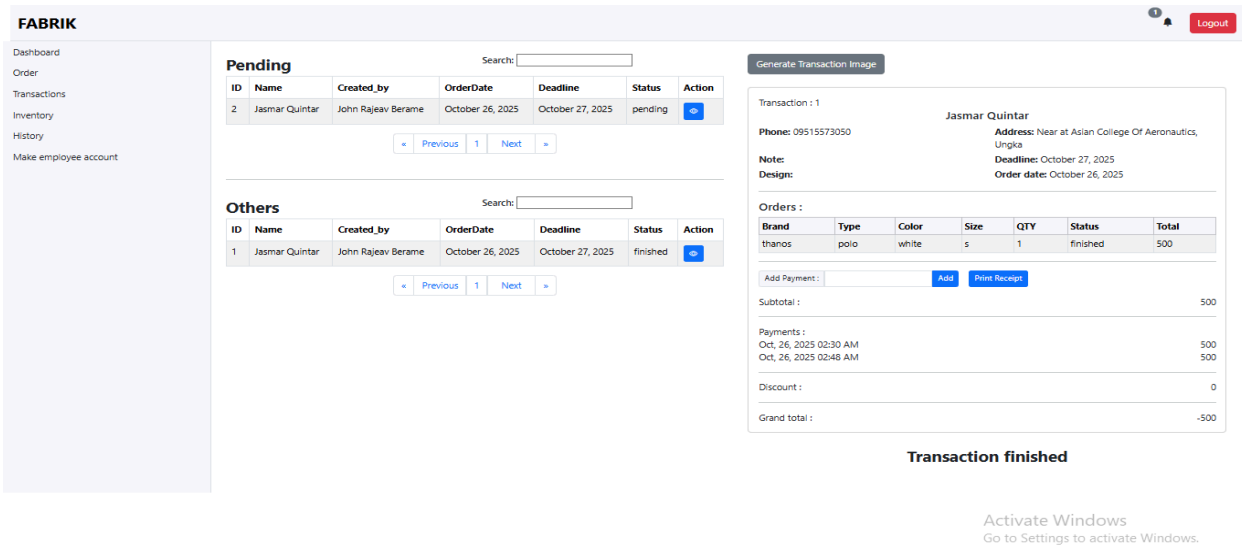
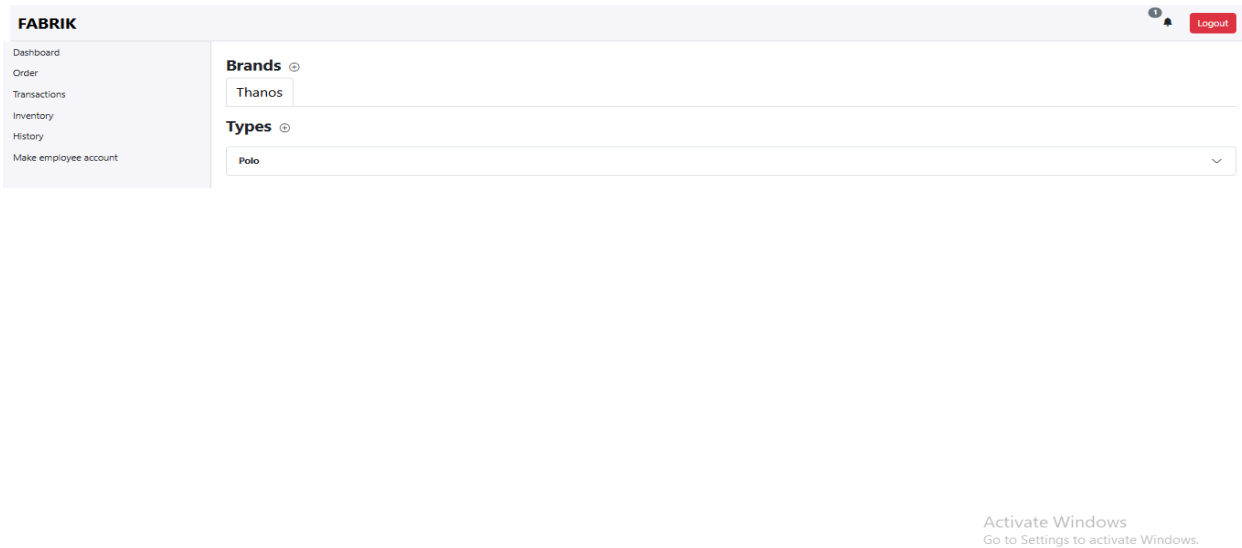


Figure 5

Inventory Management Page



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Figure 5.1

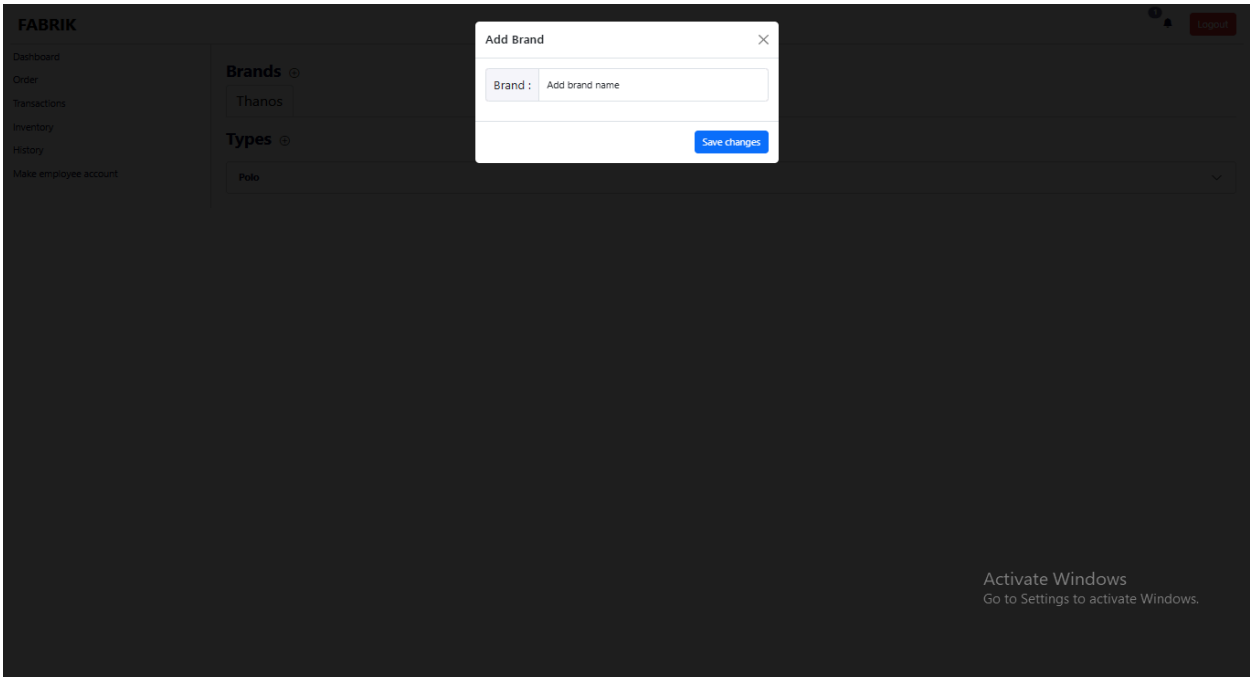


Figure 5.2

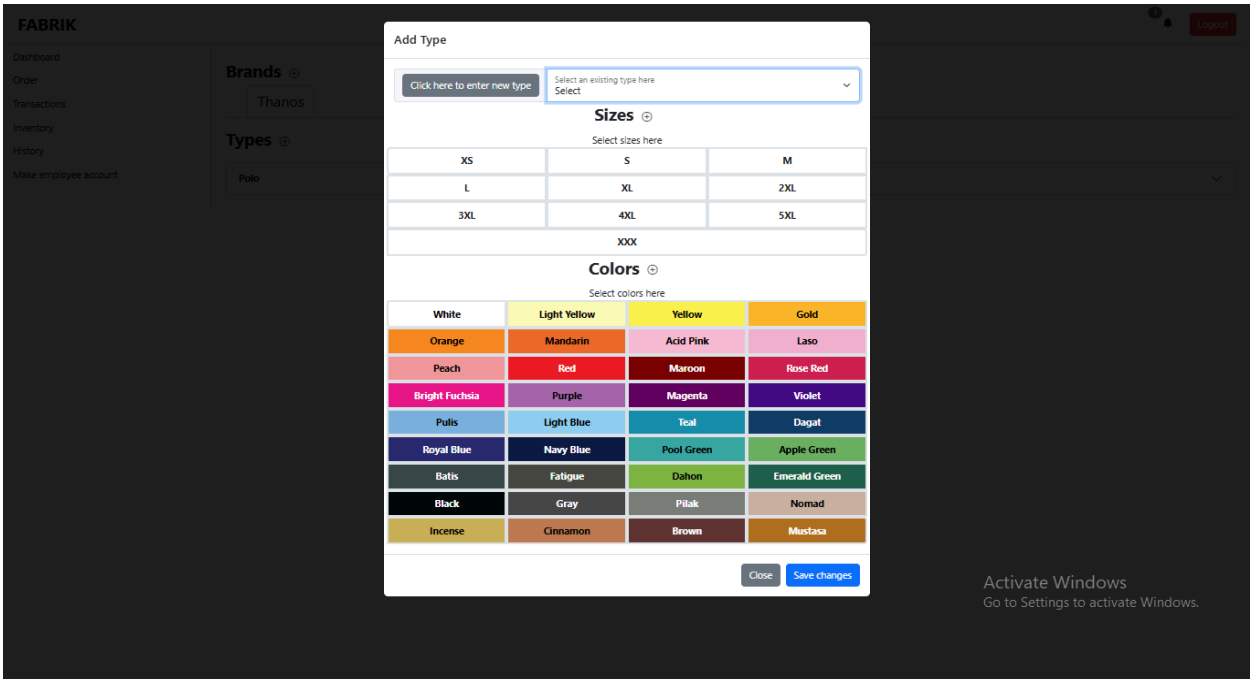


Figure 5.3

Figure 5.4[illegible]

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Figure 5.5

FABRIK

Dashboard

Order

Transactions

Inventory

History

Make employee account

Brands

Thanos

Types

Polo

Add new stocks

Return item

Delete Table

	XXX	XS	S	M	L	XL	2XL	3XL	4XL	5XL
White	0	10	0	0	0	0	0	0	0	0
Light Yellow	0	0	0	0	0	0	0	0	0	0
Yellow	0	0	0	0	0	0	0	0	0	0
Gold	0	0	0	0	0	0	0	0	0	0
Orange	0	0	0	0	0	0	0	0	0	0
Mandarin	0	0	0	0	0	0	0	0	0	0
Acid Pink	0	0	0	0	0	0	0	0	0	0
Liso	0	0	0	0	0	0	0	0	0	0
Peach	0	0	0	0	0	0	0	0	0	0
Red	0	0	0	0	0	0	0	0	0	0
Maroon	0	0	0	0	0	0	0	0	0	0
Rose Red	0	0	0	0	0	0	0	0	0	0
Bright Fuchsia	0	0	0	0	0	0	0	0	0	0
Purple	0	0	0	0	0	0	0	0	0	0
Magenta	0	0	0	0	0	0	0	0	0	0
Violet	0	0	0	0	0	0	0	0	0	0
Pulls	0	0	0	0	0	0	0	0	0	0
Light Blue	0	0	0	0	0	0	0	0	0	0
Teal	0	0	0	0	0	0	0	0	0	0
Dagat	0	0	0	0	0	0	0	0	0	0
Royal Blue	0	0	0	0	0	0	0	0	0	0
Navy Blue	0	0	0	0	0	0	0	0	0	0

Figure 5.6

FABRIK

Dashboard

Order

Transactions

Inventory

History

Make employee account

Brands

Thanos

Types

Polo

Add new stocks

Return item

Delete Table

	XXX	XS	S	M	L	XL	2XL	3XL	4XL	5XL
White	0	10	0	0	0	0	0	0	0	0
Light Yellow	0	0	0	0	0	0	0	0	0	0
Yellow	0	0	0	0	0	0	0	0	0	0
Gold	0	0	0	0	0	0	0	0	0	0
Orange	0	0	0	0	0	0	0	0	0	0
Mandarin	0	0	0	0	0	0	0	0	0	0
Acid Pink	0	0	0	0	0	0	0	0	0	0
Liso	0	0	0	0	0	0	0	0	0	0
Peach	0	0	0	0	0	0	0	0	0	0
Red	0	0	0	0	0	0	0	0	0	0
Maroon	0	0	0	0	0	0	0	0	0	0
Rose Red	0	0	0	0	0	0	0	0	0	0
Bright Fuchsia	0	0	0	0	0	0	0	0	0	0
Purple	0	0	0	0	0	0	0	0	0	0
Magenta	0	0	0	0	0	0	0	0	0	0
Violet	0	0	0	0	0	0	0	0	0	0
Pulls	0	0	0	0	0	0	0	0	0	0
Light Blue	0	0	0	0	0	0	0	0	0	0
Teal	0	0	0	0	0	0	0	0	0	0
Dagat	0	0	0	0	0	0	0	0	0	0
Royal Blue	0	0	0	0	0	0	0	0	0	0
Navy Blue	0	0	0	0	0	0	0	0	0	0

Delete Table?

NoYes

Activate Windows
Go to Settings to activate Windows.

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Figure 6

History Page

FABRIK

[Dashboard](#)
[Order](#)
[Transactions](#)
[Inventory](#)
[History](#)
[Make employee account](#)

History

Search...

ID	Detail	Inventory ID	Brand	Type	Color	Size	Old Value	New Value	Changed Value	Date	Time	Remarks
6	Add Stock	2	thanos	polo	white	xs	0	10	10	October 26 2025	05:01 AM	Stocks added in 2

[Previous](#) [1](#) [Next](#)

Inventory

Search...

Inventory ID	Brand	Type	Color	Size	Quantity
361	thanos	polo	mustasa	xxx	No Stocks
360	thanos	polo	mustasa	5xl	No Stocks
359	thanos	polo	mustasa	4xl	No Stocks
358	thanos	polo	mustasa	3xl	No Stocks
357	thanos	polo	mustasa	2xl	No Stocks
356	thanos	polo	mustasa	xl	No Stocks
355	thanos	polo	mustasa	l	No Stocks
354	thanos	polo	mustasa	m	No Stocks
353	thanos	polo	mustasa	s	No Stocks
352	thanos	polo	mustasa	xs	No Stocks

[Previous](#) [1](#) [Next](#)

Activate Windows

Go to Settings to activate Windows.

Make Employee Account

Figure 7

FABRIK

[Dashboard](#)
[Order](#)
[Transactions](#)
[Inventory](#)
[History](#)
[Make employee account](#)

History

Search...

ID	Detail	Inventory ID	Brand	Type	Color	Size	Old Value	New Value	Changed Value	Date	Time	Remarks
6	Add Stock	2	thanos	polo	white	xs	0	10	10	October 26 2025	05:01 AM	Stocks added in 2

[Previous](#) [1](#) [Next](#)

Inventory

Search...

Inventory ID	Brand	Type	Color	Size	Quantity
361	thanos	polo	mustasa	xxx	No Stocks
360	thanos	polo	mustasa	5xl	No Stocks
359	thanos	polo	mustasa	4xl	No Stocks
358	thanos	polo	mustasa	3xl	No Stocks
357	thanos	polo	mustasa	2xl	No Stocks
356	thanos	polo	mustasa	xl	No Stocks
355	thanos	polo	mustasa	l	No Stocks
354	thanos	polo	mustasa	m	No Stocks
353	thanos	polo	mustasa	s	No Stocks
352	thanos	polo	mustasa	xs	No Stocks

[Previous](#) [1](#) [Next](#)

Activate Windows

Go to Settings to activate Windows.

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Curriculum Vitae



Name: JASMAR I, QUINTAR

Address: Brgy. Ungka Jaro Iloilo City, Zone 3, Old railway rd.

Sex: male

Father: GERONIMO I, QUINTAR

Mother: MARY GRACE I, QUINTAR

Elementary School: Sambag Elementary School (2007-2013)

Junior High School: Pavia National High School (2013-2017)

**Senior High School: Pavia National High School (HUMSS Strand
2017- 2019)**

Collage: University of Iloilo (SY 2021-2026)

PHINMA UNIVERSITY OF ILOILO
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Name: JOHN RAJEAV J, BERAME

Address:

Sex: Male,agi

Father: RENANTE J, BERAME

Mother: JANETH J, BERAME

Elementary School:

Junior High School:

Senior High School:

Collage: University of Iloilo (SY 2021-2026)

PHINMA UNIVERSITY OF ILOILO
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Grammarian Certificate

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MEMORANDUM OF AGREEMENT (MOA)

Capstone Project Collaboration

This Memorandum of Agreement is made and entered into this ____ day of _____, 20____, by and among:

- **PHINMA University of Iloilo – College of Information Technology Education, represented by the Capstone Project Adviser and the Dean, hereinafter referred to as the Institution;**
 - **[Client Name], located at [Client Address], hereinafter referred to as the Client;**
 - **[Student Name/s], enrolled in the Capstone Project course, hereinafter referred to as the Student/s.**
-

1. PURPOSE

This agreement formalizes the collaboration between the Student/s and the Client for the implementation of the Capstone Project entitled:

Project Title: Fabrikflow POS and Inventory Management System with SMS Notification for Custom T-Shirt Printing

The project aims to deliver a real-world solution aligned with the Client's operational needs and the academic requirements of the Institution.

2. ROLES AND RESPONSIBILITIES

A. The Student/s shall:

- **Apply the approved Capstone Project within the Client's operational environment.**
- **Maintain professionalism, integrity, and confidentiality throughout the engagement.**
- **Submit progress reports, final documentation, and all academic deliverables to the Institution.**
- **Ensure the system is deployed and fully tested within three (3) months from the date of MOA signing.**

B. The Client shall:

- **Provide access to necessary resources, data, and personnel to support project implementation.**
- **Collaborate with the Student/s during development, testing, and evaluation phases.**
- **Participate in the system testing and provide formal feedback using the approved evaluation rubric.**
- **Coordinate with the Institution regarding the final defense schedule.**

C. The Institution shall:

- **Supervise and monitor the progress and quality of the Capstone Project.**
 - **Ensure compliance with academic standards and institutional policies.**
 - **Facilitate communication and coordination between the Student/s and the Client.**
-

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3. DURATION

This agreement shall be effective from _____ to _____, unless extended or terminated by mutual written consent of all parties.

4. CONFIDENTIALITY

All parties agree to maintain the confidentiality of sensitive information shared during the course of the project. No data shall be disclosed to third parties without prior written consent.

5. TERMINATION

This agreement may be terminated by any party with written notice, subject to review and approval by the Institution. All parties shall ensure proper turnover and documentation in the event of termination.

6. FINAL DEFENSE AND ACADEMIC STANDING

- The final defense shall be scheduled on February 20, 2026, or on a mutually agreed date between the Client and the Capstone Panelists.
 - The Student/s shall receive an “INC” (Incomplete) grade for the first semester until the system is successfully deployed and verified.
 - Failure to comply with the deployment and documentation requirements shall result in the conversion of the “INC” grade to a failing mark (“5.0”).
-

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7. SIGNATORIES

By signing below, all parties acknowledge and agree to the terms and conditions of this Memorandum of Agreement.

For the Institution

Seth dA. Nono, MSCS

Dean, College of Information Technology Education

Date: _____

KRISLYN MAE D. SINOY

Capstone Project Adviser,

Date: _____

PHINMA UNIVERSITY OF ILOILO
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

For the Client

KEN DIVINAGRACIA

Authorized Representative

Date: _____

For the Student/s

JASMAR I, QUINTAR

Signature

Date: _____

JOHN RAJEAVE J, BERAME

Signature

Date: _____

.

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